

Review

The Impact of AI and LMS Integration on the Future of Higher Education: Opportunities, Challenges, and Strategies for Transformation

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Abstract: The integration of artificial intelligence (AI) and learning management systems (LMS) is revolutionising higher education, offering unprecedented opportunities for personalised learning, adaptive assessments, and data-driven decision-making. This review investigates the impact of AI–LMS integration on educational quality, student success, and institutional performance in higher education. In addition, this review not only examines the technological integration but also evaluates how AI–LMS systems contribute to sustainable development in higher education through reduced resource consumption, improved accessibility, and enhanced educational equity. Following the PRISMA 2020 guidelines, a comprehensive search of the Scopus database yielded 60 relevant studies published between 2014 and 2023. The review reveals significant benefits of AI–LMS integration, including enhanced student engagement, personalised learning paths, and improved learning outcomes. Key applications include AI-powered conversational agents, adaptive assessments, and learning analytics. However, challenges such as data privacy concerns, algorithmic bias, and the need for faculty training were also identified. The findings highlight strategies for effective AI–LMS implementation, emphasising the importance of ethical considerations and addressing the digital divide. Results demonstrate that AI–LMS integration can significantly enhance educational quality and student performance when implemented thoughtfully. The review also uncovers areas requiring further research, including long-term impacts on learning outcomes, scalability of AI–LMS solutions, and strategies for ensuring equitable access. Future studies should focus on longitudinal assessments of AI–LMS effectiveness, the development of ethical frameworks for AI in education, and the exploration of AI–LMS applications in diverse educational contexts. This review provides valuable insights for higher education institutions seeking to leverage AI–LMS integration to transform teaching and learning practices.

Keywords: artificial intelligence; learning management systems; higher education; educational technology; personalised learning



Citation: Alotaibi, N.S. The Impact of AI and LMS Integration on the Future of Higher Education: Opportunities, Challenges, and Strategies for Transformation. *Sustainability* **2024**, *16*, 10357. <https://doi.org/10.3390/su162310357>

Academic Editor: Aras Bozkurt

Received: 23 September 2024

Revised: 4 November 2024

Accepted: 4 November 2024

Published: 27 November 2024



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1. Introduction

The influence of artificial intelligence (AI) on higher education has been profound, revolutionising teaching, learning, and the operations of institutions [1]. AI-powered tools and systems have revolutionised education by providing personalised learning experiences, adaptive assessments, and data-driven decision-making [2]. These advancements have significantly improved educational quality and student success, as demonstrated [3,4]. Additionally, the integration of AI and LMS in higher education aligns strongly with sustainable development goals, particularly in promoting quality education (SDG 4) and responsible consumption (SDG 12) [5]. This technology combination offers opportunities to significantly reduce the environmental impact of educational delivery through digitalisation of learning materials, optimisation of resource usage, and reduction in physical infrastructure requirements [6]. Furthermore, AI-powered LMS solutions support social sustainability by democratising access to quality education and enabling flexible learning

models that accommodate diverse student needs and circumstances [7]. The economic sustainability dimension is addressed through more efficient resource allocation, reduced operational costs, and improved scalability of educational programs [8].

Furthermore, the combination of AI and learning management systems (LMS) has paved the way for the creation of intelligent tutoring systems, chatbots, and recommendation engines [9]. These tools offer personalised assistance to students and enhance their level of engagement [2,10]. In addition, AI has enhanced administrative processes in higher education, including admissions, student advising, and resource allocation, resulting in improved efficiency [10] and cost-effectiveness [11]. However, the integration of AI in higher education has sparked ethical debates surrounding issues such as data privacy, algorithmic bias, and fairness [3,12]. Institutions face the task of addressing these challenges while also prioritising accessibility, equity, and faculty preparedness [13].

However, the combination of artificial intelligence (AI) and learning management systems (LMS) has a profound effect on the delivery and experience of education in higher education [14]. Furthermore, AI is having a significant impact on the LMS industry, bringing a fresh wave of customisation, automation, and data-driven analysis. Researchers acknowledged the impressive capabilities of AI, such as curating personalised learning paths, suggesting relevant materials based on individual progress [15], and offering immediate feedback through automated grading and analysis [16]. AI-powered conversational agents and recommendation systems in LMS have significantly enhanced student engagement and offered personalised support [2,10]. In addition, AI's capabilities, such as personalised learning and data-driven decision-making, address the current challenges faced by higher education institutions, including scalability, student engagement, and resource management [17]. This connection would underscore the necessity and timeliness of integrating AI into LMS, aligning with the evolving demands of modern education systems [18].

Despite the integration of AI-powered LMS in higher education offering promising advancements, notable challenges and ethical considerations must be addressed. Concerns regarding data privacy, security, algorithmic bias, and fairness cannot be overlooked [3,19]. This is particularly important when considering the digital divide and ensuring equal access for all students [13]. Additionally, sufficient training and support for faculty members is crucial for successfully adopting these new technologies. This study's focus on AI-LMS integration is particularly timely given the growing emphasis on sustainable development in higher education. By examining how these technologies can support both educational excellence and sustainability goals, this research contributes to the broader discussion of technology's role in creating more sustainable educational models. The findings will help institutions balance the imperatives of educational innovation with environmental responsibility and social inclusivity.

From the above discussion, there is a need for a review that can illustrate what is necessary to bridge the gap between faculty and AI-powered LMS and highlight the importance of research in this area. This study aims to add to the expanding corpus of research by investigating the potential impact and benefits of AI and LMS integration on teaching and learning practices, administrative operations, and the overall transformation of the higher education landscape. The following questions are addressed:

1. What are the key benefits and opportunities associated with integrating AI and LMS in higher education institutions?
2. What are the major challenges and ethical concerns that need to be addressed when implementing AI-powered LMS in higher education?
3. How can higher education institutions effectively leverage the integration of AI and LMS to enhance educational quality, student success, and overall institutional performance?

This will ultimately guide higher education institutions towards a more effective and responsible integration, ensuring a transformed learning landscape that benefits all students and faculty.

2. Materials and Methods

This study's literature review methodology is based on the PRISMA Statement 2020 [20]. The main purpose was to investigate how artificial intelligence and big data analytics may improve higher education. The literature review approach allows for a complete examination of previous studies, considering significant factors influenced by human interactions. The procedure was divided into three sections. The first step is to utilise keywords to find relevant articles from respected sites around the world that have researched this topic [21]. An exhaustive search for relevant research articles in the Scopus database was conducted at this stage of the inquiry. A keyword pattern was used to ensure a search that is especially focused on higher education in accordance with the study strategy. The following was the keyword search query: "teaching and learning" OR "student engagement" OR "personalized learning" AND "artificial intelligence" AND "learning management systems". Following the keyword search, exclusion criteria were used with inclusion criteria throughout the search. The criteria required that the publications be from 2014 to 2023, remove any duplicate sources that were not part of the literature review, focus only on articles, book chapters, and reviews, and include only materials written in English. The predefined keyword query yielded 102 research publications that satisfied the given requirements. After examining the 102 research publications, it was determined that they were pertinent to the study subject. A further examination identified 82 publications that provided a response to the study inquiry. Following a comprehensive linguistic examination, a total of 60 documents were selected for further investigation. Figure 1 displays the specific information on the extraction process.

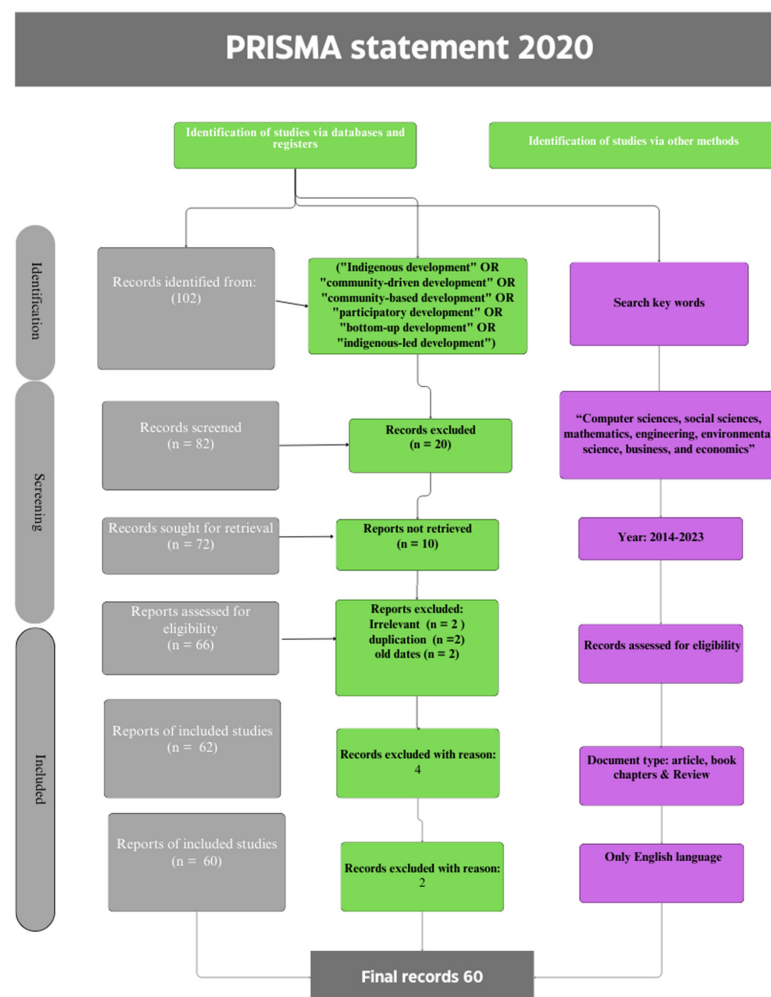


Figure 1. Inclusion and exclusion criteria of PRISMA Statement 2020.

Inclusion Criteria:

- Publications from 2014 to 2023
- Articles, book chapters, and reviews
- Materials written in English

Exclusion Criteria:

- Duplicate sources that were not part of the systematic literature review
- Publications outside the date range of 2014 to 2023
- Document types other than articles, book chapters, and reviews
- Non-English publications

3. Results

To explore the convergence of artificial intelligence (AI) and learning management systems (LMS) in higher education, Table 1 provides helpful data. The data span 2015–2023, showing that the study’s research objective focuses on recent and present achievements in this sector. The study examined the numerous aspects of integrating artificial intelligence (AI) with learning management systems (LMS) in higher education using 52 journals, books, and other publications. Due to the 16.53% annual growth rate in publications, this sector is seeing more research, making the study goal relevant and contemporary.

Table 1. Main information.

Description	Results
Timespan	2015:2023
Sources (journals, books, etc.)	52
Documents	60
Annual growth rate %	16.53
Document average age	3.57
Average citations per doc	7.672
References	2071
DOCUMENT CONTENTS	
Keywords plus (ID)	290
Author’s keywords (DE)	234
AUTHORS	
Authors	3
Authors of single-authored docs	0
Single-authored docs	0
Co-authors per doc	3
International co-authorships %	16.39
Article	16
Book	1
Book chapter	11
Conference paper	31
Review	2

The study included publications with an average age of 3.57 years, indicating its use of current research. This is crucial to understanding higher education AI and LMS usage now and in the future. The average of 7.672 citations per document implies that the selected

literature has a significant academic impact, supporting the research purpose. The study's 2071 references demonstrate its broad scope and ensure a thorough assessment.

The study comprised conference papers, journals, and book chapters to cover all aspects of AI and LMS in higher education research and application. The study is multidisciplinary and worldwide, with an average of three co-authors per document and 16.39% international co-authorships. This helps understand the complex effects of merging AI and LMS in higher education.

In addition, Figure 2 shows how article distribution over time supports the study's goal of examining higher education's usage of AI and LMS. The annual publishing count fluctuates, with a recent increase. In 2016, there was just one publication, showing this topic was still relatively new. In 2017 and 2021, the number of articles increased to 5 and 12, respectively, showing increased interest and study.

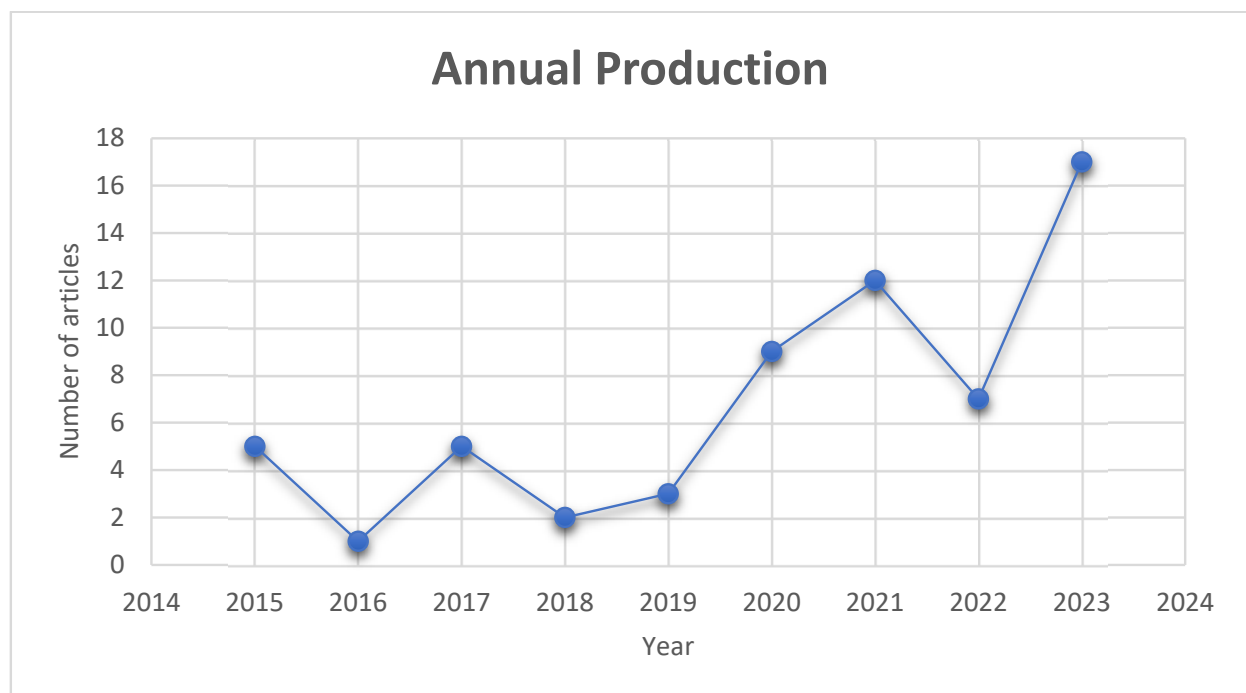


Figure 2. Annual production of articles.

In 2023, 17 articles showed a rise in academic interest in AI–LMS integration in higher education. Due to its timeliness and relevance, this rise supports the study's goal. The inquiry seeks to understand the latest developments and trends in this ever-changing industry. However, the drop from 12 publications in 2021 to 7 in 2022 may indicate a short slowdown or shift in study focus. Due to new technologies and instructional techniques, the publishing pattern changes constantly. This requires constant research and modification.

Furthermore, Figure 3 shows the distribution of publications across multiple sources. The results show that the articles come from a diverse range of academic journals, conference proceedings, and book series. The prominence of “Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)” with three articles and “ACM International Conference Proceeding Series” with two articles suggests that the research has thoroughly investigated the technical and computational aspects of integrating AI and LMS in higher education. These prominent sources are known for their focus on cutting-edge technology, computer science, and artificial intelligence, which aligns well with the study's aim.

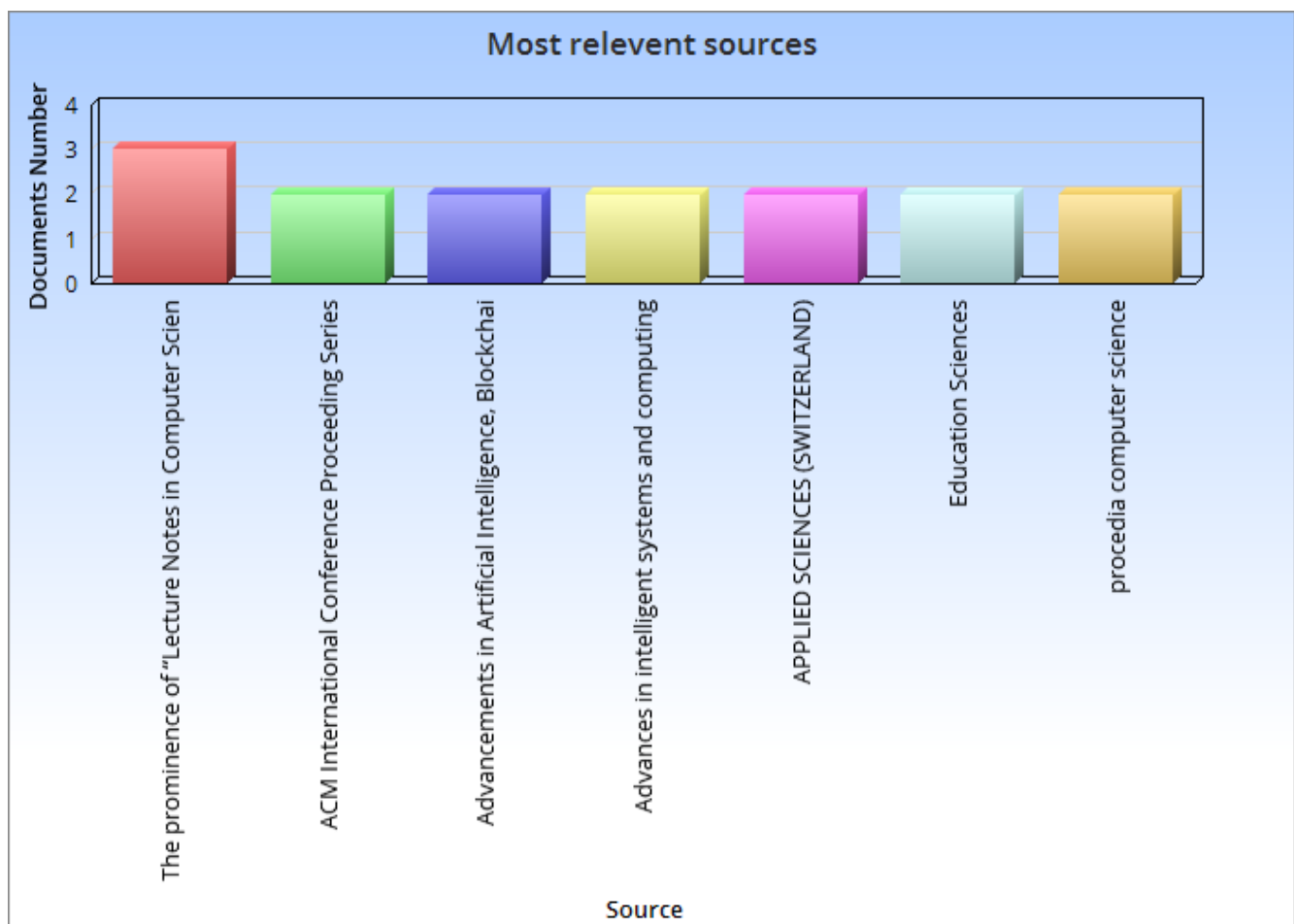


Figure 3. Most relevant sources for data collection.

In addition, the inclusion of sources like “Advancements in Artificial Intelligence, Blockchain Technology, and IoT in Higher Education: Mitigating the Impact of COVID-19” and “Education Sciences” indicates that the study has also considered the broader educational and social implications of AI and LMS adoption in higher education, particularly in the context of the COVID-19 pandemic and its impact on teaching and learning. Moreover, the concentration of articles in certain sources, such as the two articles in “Applied Sciences (Switzerland)” and “Procedia Computer Science”, suggests that these venues are prominent in publishing high-quality research on the intersection of AI, technology, and higher education. This information can guide future researchers in identifying the most relevant and impactful sources for their work in this field.

The bibliometric analysis in this study was conducted using the bibliometrix package in RStudio, an integrated development environment for the R programming language [22]. The bibliometrix package is a comprehensive tool for quantitative research in scientometrics and bibliometrics, providing a wide range of functions for analysing bibliographic data [23].

Figure 4 illustrates the relationships between universities, countries, and source materials through a three-field plot. The research highlights several universities, including Muscat College, Jazan University, and several institutions from Brazil and Australia, on the left side of the plot (AU_UN). Consistent with the research objective, this signifies that the study investigated the implementation of AI and LMS in a wide variety of higher education establishments. In addition, the primary focus of the narrative (AU_CO) is on the source materials, comprising book chapters, conference proceedings, and journal articles. Based on the scholarly articles “Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)”

and “Advancements in Artificial Intelligence, Blockchain Technology, and IoT in Higher Education: Mitigating the Impact of COVID-19”, it is evident that the researchers have conducted extensive research on the educational, computational, and technical dimensions of integrating learning management systems (LMS) with artificial intelligence (AI) in higher education.

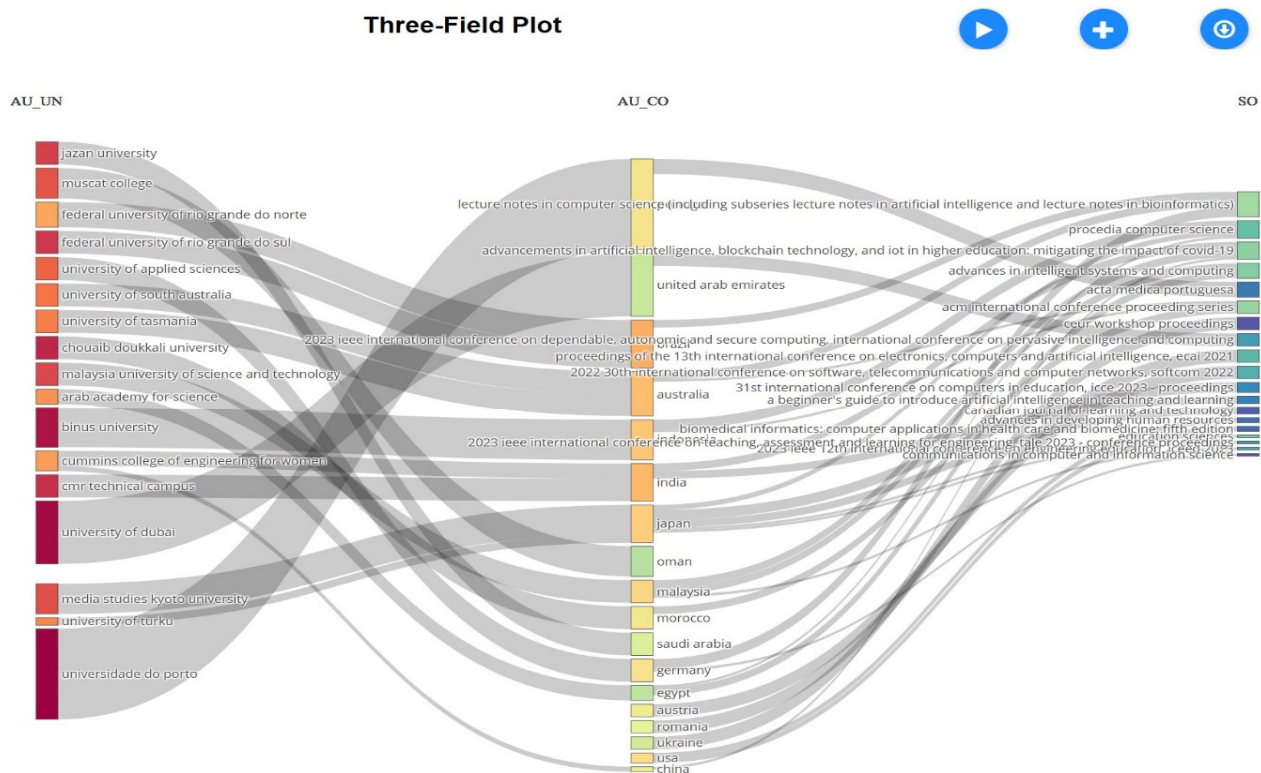


Figure 4. Three-plot analysis between universities, countries, and sources.

Furthermore, the countries linked to the research are displayed on the right side of the plot (SO). These countries encompass both developed nations, such as the United States and Germany, and emerging economies, like Saudi Arabia and Malaysia. The extensive geographic coverage of this study suggests that it investigated the integration of learning management systems (LMS) and artificial intelligence (AI) in higher education on a global scale. This strengthens the research objective and contributes to a comprehensive understanding of the subject matter.

Furthermore, the co-word network analysis in Table 2 provides significant insight into fundamental ideas and their interconnections within the network. The nodes with the greatest betweenness centrality are “artificial intelligence”, “learning management system”, and “learning systems”. This suggests that these concepts play a vital role in connecting different topics and ideas in the study area. This is consistent with the study’s purpose since these core ideas are essential for comprehending the merging of artificial intelligence (AI) and learning management systems (LMS) in higher education.

The significant proximity centrality of “learning systems”, “students”, and “e-learning” indicates that these nodes have strong connections to the larger network, emphasising their significance in the wider context of integrating AI with LMS. This is important because the research seeks to investigate the influence of these technologies on the process of teaching, learning, and student involvement in higher education. The PageRank centrality measurements reveal that “learning systems”, “artificial intelligence”, and “learning management systems” are the most highly rated nodes, highlighting their significance and the enormous research attention devoted to these fundamental topics. This connection with the

research purpose enhances the study's capacity to provide a thorough comprehension of the incorporation of AI and LMS in higher education.

Table 2. Co-word network analysis.

Node	Betweenness	Closeness	Page Rank
artificial intelligence	193.105	0.016	0.092
learning management system	138.771	0.016	0.086
learning systems	269.187	0.017	0.097
students	188.876	0.017	0.083
e-learning	117.283	0.015	0.078
teaching	21.36	0.014	0.05
computer-aided instruction	3.983	0.013	0.039
education	49.228	0.013	0.036
teaching and learning	3.857	0.012	0.033
education computing	3.375	0.013	0.031
information management	5.642	0.013	0.032
deep learning	0	0.011	0.012
personalised learning	0	0.011	0.012
active learning	0.479	0.012	0.015
curricula	0.11	0.011	0.012
data mining	0	0.012	0.011
online learning	0.029	0.012	0.014
online systems	0.542	0.012	0.013
blended learning	0.086	0.011	0.01
COVID-19	0	0.011	0.008
e-learning management systems	0.018	0.011	0.01

The use of co-word network analysis and thematic mapping helps to understand the specific impacts of AI on learning management systems (LMS) [24]. These tools are effective for uncovering trends and patterns within the research landscape, but providing a more in-depth explanation of their contributions and comparing them to other bibliometric techniques, such as citation analysis, can strengthen the methodological foundation and offer a more comprehensive understanding of the field [25].

Furthermore, Figure 5 displays a co-occurrence network that provides a thorough visual depiction of the interactions and interconnections among major ideas in the field of artificial intelligence and learning research. The main emphasis of the study is highlighted by the significant “artificial intelligence” and “learning” nodes in the heart of the network, which correspond with the broader aims. The complex network of links stemming from these fundamental notions illustrates the diverse and complicated character of the discipline, highlighting significant sub-topics and related themes like “student”, “computer”, and “data”. The extensive amount of intricacy and interdependence implies that the study has thoroughly explored the intricate relationships between artificial intelligence technologies and their utilisation in educational settings.

Furthermore, the intricate arrangement of the network, characterised by its compact clusters and branching paths, suggests that the study has thoroughly examined a wide array of viewpoints and methodologies within this field. The visual intricacy reflects the underlying difficulties and multifaceted nature of incorporating artificial intelligence into educational systems and contexts. Through a rigorous examination of this network, researchers can acquire valuable knowledge about the present condition of the field, detect emerging patterns, and reveal potential subjects for future research that can enhance the comprehension of artificial intelligence's transformative capacity in higher education.

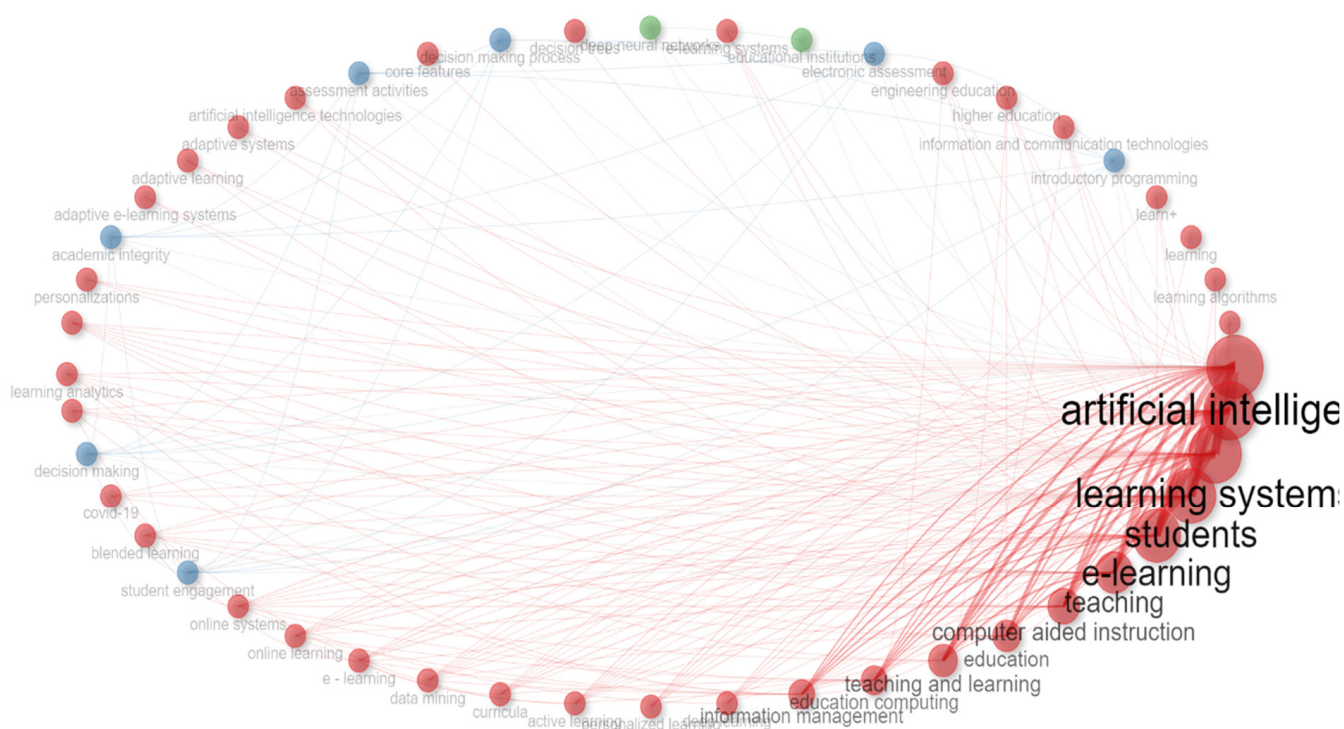


Figure 5. Co-occurrence network.

Finally, Figure 6 showcases a thematic map that offers a comprehensive visual depiction of the principal themes and their interconnections as they pertain to the body of research concerning the amalgamation of learning management systems (LMS) and artificial intelligence (AI) in the context of higher education. By dividing the map into four quadrants according to density and centrality metrics, the map provides insightful information regarding the structure and dynamics of the research landscape. The “Motor Themes”, evident in the upper-right quadrant, are enduring and influential subjects that significantly impact the research domain. The themes “learning systems”, “e-learning”, “learning management system”, and “teaching and learning” predominate in this quadrant and the “Basic Themes” quadrant. The investigation of the convergence of learning management systems (LMS) and artificial intelligence (AI) in higher education is in accordance with these fundamental ideas, which are not only interrelated but also pivotal to the research.

In addition, the research is organised around “Niche Themes”, which are isolated yet highly developed topics in the upper-left quadrant. This domain comprises the concepts of “students” and “student engagement”, indicating that although these are significant focal points, they might not be as thoroughly intertwined with the wider research framework. Also, the “Basic Themes”, located in the lower-right quadrant, are the foundational yet less-expanded concepts that serve as the basis for the research domain. The theme of “artificial intelligence” is present near this quadrant, signifying that although AI is a fundamental component of research, additional investigation and integration with other significant themes are still possible.

In conclusion, the lower-left quadrant comprises the “Emerging or Declining Themes”, which are the periphery and least developed subjects that might signify nascent trends or domains that require additional inquiry. The sparseness of content in this quadrant indicates that the research has remained firmly centred on the fundamental concepts associated with integrating AI and LMS in higher education. Through analysis of the thematic map, researchers can ascertain regions that warrant further investigation, discern the interdependence and comparative significance of various themes, and acquire significant insights into the present condition of the field.

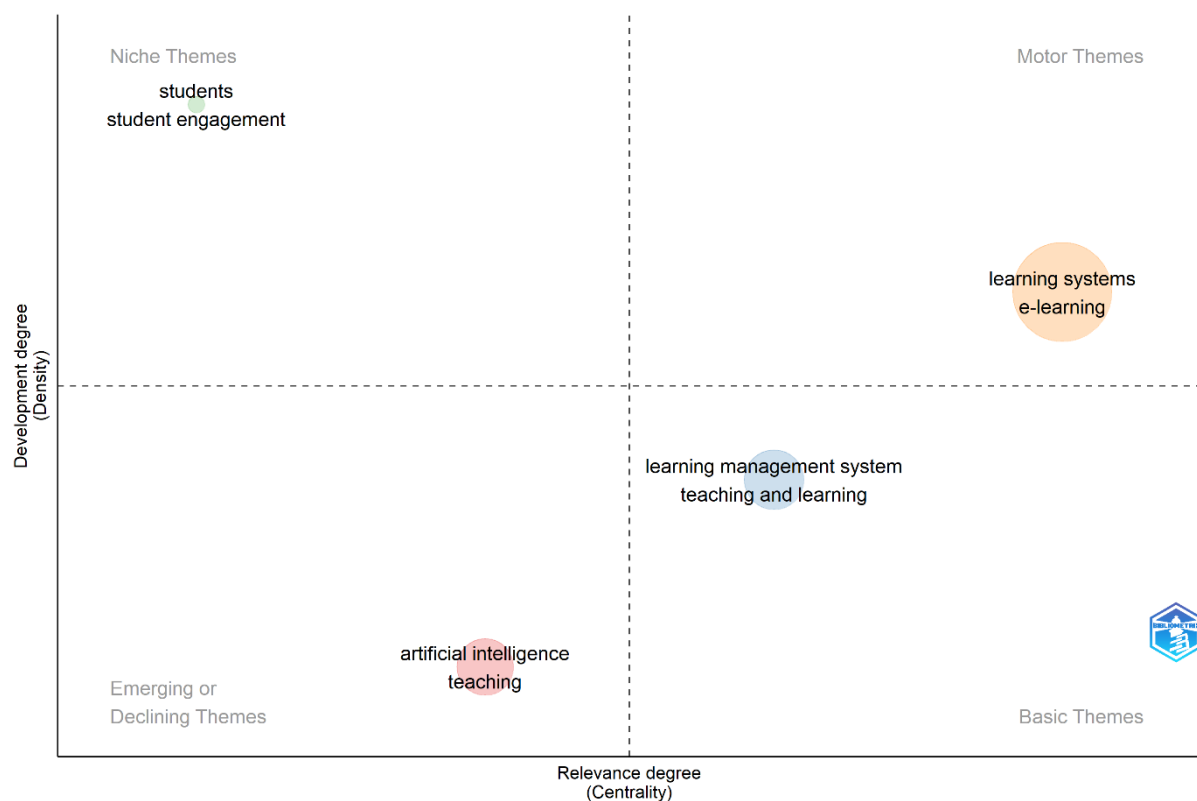


Figure 6. Thematic map of the key themes and their relationships in the research on integrating artificial intelligence (AI) and learning management systems (LMS) in higher education.

4. Discussion

4.1. Benefits and Opportunities Associated with Integrating AI and LMS in Higher Education Institutions

In higher education, the integration of learning management systems (LMS) and artificial intelligence (AI) presents a variety of potential opportunities, challenges, and ethical concerns. According to the findings of Uunona & Goosen. [19], the use of ethical standards in AI is important for responsible teaching and learning applications in a university setting. They found that AI-powered online learning applications, when designed with ethical considerations, can improve diversity, inclusion, and student outcomes. Similarly, Atif et al. [2] investigated AI-enabled remote learning and teaching using pedagogical conversational agents and learning analytics across multiple universities. Their findings revealed that AI-enabled conversational agents and learning analytics in LMS provided personalised support, enhanced engagement and learning, and empowered students in a blended learning environment during the pandemic. In addition, shifting focus to the design of workforce training systems for the physically challenged, Hamza et al. [26] discovered that integrating AI and AR/VR into LMS facilitated personalised learning approaches and improved life skills development for this group of learners. Additionally, Karyy et al. [11] proposed a model for organising the educational process using AI technologies, which revealed that AI applications in LMS personalised learning paths, provided adaptive assessments, and generated data-driven insights to improve the overall educational process and learning outcomes.

Furthermore, Kumari et al. [27] delved into the use of AI-based digital technologies in smart universities and found that integrating AI technologies across university campuses and LMS personalised and enhanced the overall educational infrastructure and experience for students. Similarly, Kunnath et al. [10] developed an intelligent and adaptive learning management system using data mining and AI in an engineering college setting. Their findings indicated that applying AI and data mining techniques to LMS data helped

identify gaps in student knowledge and provided personalised learning paths to improve learning efficiency. Lim et al. [28] focused on enhancing LMS by integrating learning styles and adaptive courses in a university in Malaysia. They discovered that extending LMS with AI-driven adaptivity based on student learning styles enabled personalised support for both teachers and learners, thereby improving learning effectiveness. Likewise, Paba-Medina [29] designed a gamified tool for developing citizenship competencies and found that AI-powered gamification tools integrated with LMS reinforced the teaching and learning of important skills like citizenship competencies in an engaging way.

Additionally, AI-based educational platforms and tools (MOOCs, games, LMS, AR/VR, etc.) can tailor learning experiences to facilitate knowledge and skill development, yielding positive outcomes in different learning settings [3]. In a related study, Santoso et al [30] investigated the effectiveness of LMS digital tools used by academics to foster student engagement and revealed that 90% of academics agreed that AI-powered tools in LMS, such as discussion forums, chatbots, and assessment features, encouraged interactivity and student engagement in online learning. Also, integrating AI into LMS for learning analytics and personalised learning enabled early identification of at-risk students and provided data-driven insights for teachers to enhance student education, with initial testing showing promising results [31]. Similarly, Berry [32] applied AI to adaptive learning environments and developed an AI-based adaptive e-learning system in LMS that personalised instruction based on student learning style and knowledge level assessed using the C4.5 decision tree algorithm, leading to improvements in student knowledge levels.

Also, AI integration in learning management systems (LMS) can lead to personalised learning and adaptive assessments. These opportunities align with current AI advancements in education [3], but providing more practical examples of their implementation within LMS platforms can strengthen the argument for their feasibility and impact. For instance, Kunnath et al. [10] demonstrated how AI and data mining techniques applied to LMS data can identify student knowledge gaps and provide personalised learning paths, improving learning efficiency. Table 3 illustrates the studies that discuss the opportunities for AI integration in LMS across higher educational settings.

Table 3. Opportunities for AI integration in LMS across higher educational settings.

Authors and Year	Research Focus	Research Settings	Key Findings
Uunona & Goosen [19]	AI ethics for responsible teaching and learning	University setting	When ethically built, AI-powered online learning apps may promote diversity, inclusivity, and student results.
Atif et al. [2]	Teaching remotely using AI-enabled conversational bots and learning analytics	Multiple universities	AI-enabled conversational agents and learning analytics in LMS empowered students in blended learning.
Karyy et al. [11]	AI-organised educational process model	AI applications in LMS personalised learning	AI in LMS personalised learning routes, adaptive assessments, and data-driven insights improved education and results.
Kumari et al. [27]	AI-based digital technologies in smart universities	AI technologies integrated	AI technology on university campuses and in LMS personalised and improved student instruction.
Kunnath et al. [10]	Intelligent and adaptive learning management system using data mining and AI	Engineering college	AI and data mining on LMS data identified student knowledge gaps and personalised learning routes to boost learning efficiency.

Table 3. Cont.

Authors and Year	Research Focus	Research Settings	Key Findings
Lim et al. [28]	Enhancement of LMS by integrating learning styles and adaptive courses	University in Malaysia	LMS with AI-driven adaptivity based on student learning styles improved learning effectiveness by personalising assistance for instructors and students.
Paba-Medina et al. [29]	Design of a gamified tool for developing citizenship competencies	AI-powered gamification tools integrated with LMS	AI-powered gamification techniques in LMS made citizenship skills learning fun.
Qushem et al. [3]	Multimodal technologies in precision education	Multiple research examples	AI-based educational platforms and tools improved knowledge and skill development in many learning contexts.
Santoso et al. [30]	Effectiveness of LMS digital tools used by academics to foster student engagement	Tshwane University of Technology	Around 90% of academics said AI-powered LMS technologies, including discussion forums, chatbots, and evaluation features, increased student participation in online learning.
Sharif et al. [31]	AI-powered learning analytics architecture for enhancing student education	Integrating AI into LMS for learning analytics	Initial testing showed that integrating AI into LMS for learning analytics and personalised learning identified at-risk kids and provided instructors with data-driven insights to improve student education.
Tendulkar et al. [33]	AI-based approaches for teaching and learning to develop critical thinking and problem solving	AI applications in LMS	AI applications in LMS helped provide focused curriculum and learning paths for students to develop into independent learners with strong thinking and problem-solving abilities.
Zagorskis et al. [34]	Learning analytics architecture for machine learning algorithm integration in LMS	Riga Technical University	Analysis of student behavioural data led to an AI-driven dynamic and personalised adaptive learning system in LMS to improve learning environments.

Moreover, Tendulkar et al. [33] explored AI-based approaches for teaching and learning to develop critical thinking and problem-solving skills. They discovered that AI applications in LMS helped provide focused curriculum and learning paths for students to develop into independent learners with strong thinking and problem-solving abilities. Finally, Zagorskis et al. [34] developed a learning analytics architecture for machine learning algorithm integration in LMS at Riga Technical University. They found that an AI-driven dynamic and personalised adaptive learning system in LMS, developed by analysing learner behavioural data, enabled more effective learning environments. In conclusion, the significant benefits and opportunities associated with integrating AI and LMS in higher education institutions include personalised learning experiences, adaptive assessments, early identification of at-risk students, increased student engagement, and data-driven insights for educators to enhance pedagogical strategies.

4.2. Challenges and Ethical Concerns Implementing AI-Powered LMS in Higher Education

In contrast to the previously discussed studies, several researchers have instead examined the difficulties and ethical considerations associated with artificial intelligence in higher education. The use of ethical criteria in AI to promote responsible teaching and learning in a university environment is important [35]. The challenges include issues such as safeguarding data privacy and security, addressing algorithmic bias and ensuring fairness, grappling with ethical concerns related to student autonomy and consent, bridging the digital gap and promoting accessibility, and providing enough faculty training and uptake [19]. Similarly, ethical considerations and privacy concerns need to be carefully navigated to ensure the responsible implementation of AI technologies in educational contexts [3]. The discussion of challenges and ethical concerns, particularly around data privacy and algorithmic bias, is indeed a crucial aspect of our work [13,19].

For example, Rerhaye et al. [13] specifically examined assessment techniques for an AI-enhanced learning management system (LMS) and pinpointed obstacles such as the need for privacy-preserving measures and effective handling of diverse data when incorporating AI into LMS. According to Santoso et al. [30], one of the challenges is the requirement for instructional design assistance in choosing suitable AI-powered tools within the LMS to guarantee both effectiveness and ethical usage.

Additionally, research into virtual human resource development, online education, and learning [36] revealed obstacles including the intricacy of virtual environments, the requirement for both structured and unstructured learning, and the socio-cultural characteristics of extensive digital networks and learning management systems. The researchers emphasised ethical concerns such as privacy invasion and the potential for biased results. In addition, Tendulkar et al. [33] examined AI-driven methodologies in education to foster the growth of critical thinking and problem-solving abilities. They emphasised the importance of meticulously crafting AI applications within learning management systems (LMS) to ensure targeted curricula and learning trajectories while addressing ethical considerations. In a similar vein, love et al. [37] investigated the requirements of adult learners and the incorporation of technology to facilitate successful online education, with a particular focus on ethical implications, such as the necessity to foster personalised learning while safeguarding privacy and potential biases in AI-powered tools integrated with learning management systems. Table 4 illustrates the studies discussing the challenges in AI-powered learning management systems.

Table 4. Challenges in AI-powered learning management systems (LMS).

Authors and Year	Research Focus	Research Settings	Key Findings
Atif et al. [2]	Remote learning and teaching using pedagogical conversational bots and learning analytics utilising AI	Multiple universities	AI-enabled conversational agents and learning analytics in LMS may boost student engagement, personalised assistance, and empowerment, improving educational quality and achievement.
Kunnath et al. [10]	Data mining and AI-based adaptive learning management systems	Engineering college	LMS conversational agents and learning analytics may improve student engagement, empowerment, and educational quality and accomplishment.
Lim et al. [28]	Integrating learning styles and adaptive courses improves LMS	University in Malaysia	Institutions may add AI-driven adaptivity to LMS based on student learning styles to provide personalised assistance.

Table 4. Cont.

Authors and Year	Research Focus	Research Settings	Key Findings
Qushem et al. [3]	Precision education using multimodal technology	Multiple research examples	AI-based educational platforms and technologies show significant potential.
Santoso et al. [30]	Academic LMS digital technology student engagement abilities	Tshwane University of Technology	Universities may promote AI-powered LMS technologies like discussion forums, chatbots, and evaluation features to boost student engagement and achievement.
Sharif et al. [31]	An AI-powered learning analytics framework to improve student learning	Learning analytics to personalise learning	Universities may use AI in LMS for learning analytics to personalise learning, identify at-risk children, and provide instructors with data-driven insights to improve student education and institutional performance.
Tendulkar et al. [33]	AI-based critical thinking and problem-solving instruction	Concentrated curriculum and learning pathways	AI applications may concentrate curriculum and learning pathways to help students become autonomous learners with strong thinking and problem-solving abilities.
Zagorskis et al. [34]	Learning analytics architecture for LMS machine learning algorithm integration	Riga Technical University	Using learner behavioural data, institutions may create AI-driven dynamic and personalised adaptive learning systems in LMS to enhance learning environments and institutional performance.
Paba-Medina et al. [29]	Citizenship competency development using gamification	AI-powered gamification tools	Institutions may use AI-powered gamification tools with LMS to engage students in citizenship skills and improve educational quality.

Furthermore, when integrating AI-powered e-book systems with LMS for personalised learning during emergency remote teaching, ref. [38] identified obstacles such as the need to address data security and privacy concerns as well as the role of e-book technology in facilitating university education during COVID-19 at a national university in Japan. Paba-Medina et al. (2020) [29] concluded with a discussion of ethical considerations regarding developing citizenship competencies through the design of a gamified tool. They emphasised the necessity of integrating LMS with AI-powered gamification tools, reinforcing the learning and instruction of critical skills while simultaneously preventing potential biases or adverse effects. In conclusion, the resolution of these ethical dilemmas and obstacles necessitates meticulous strategizing, active involvement of relevant parties, formulation of policies, and continuous assessment and appraisal to guarantee the ethical and efficient incorporation of AI into learning management systems (LMS), thereby optimising advantages for educators and students while minimising potential hazards and adverse effects.

4.3. Educational Quality, Student Success, and Overall Institutional Performance

AI-enabled conversational agents and learning analytics in LMS can provide personalised assistance, boost engagement and learning, and empower students, improving

educational quality and student performance [2]. Multiple universities employ AI-enabled remote learning and teaching, including pedagogical conversational bots and learning analytics. For example, Kunnath [10] used data mining and AI to create an intelligent and adaptable learning management system. Their results show that universities may use AI and data mining to detect student knowledge gaps and provide personalised learning routes to improve learning efficiency and student success using LMS data. Also, universities might enhance their learning management systems (LMS) by incorporating artificial intelligence (AI) technology that adapts to students' learning styles [28]. This would provide personalised assistance for both instructors and learners, leading to improved learning effectiveness and overall educational quality. Additionally, institutions can use AI-powered educational platforms and technologies such as MOOCs, games, LMS, AR/VR, etc. to customise learning experiences and promote the acquisition of information and skills, resulting in favourable results and improved educational quality [3].

Furthermore, the effectiveness of LMS digital tools used by academics to foster student engagement was explored at Tshwane University of Technology. Their findings indicate that institutions can encourage the use of AI-powered tools in LMS, such as discussion forums, chatbots, and assessment features, to increase interactivity and student engagement, contributing to improved student success [30]. Additionally, Sharif et al. (2023) [31] proposed an AI-powered learning analytics architecture for enhancing student education. They suggest that institutions can integrate AI into LMS for learning analytics to personalise learning, enable early identification of at-risk students, and provide data-driven insights for teachers to enhance student education and overall institutional performance. AI-enabled conversational agents and learning analytics in LMS improved student engagement, personalised support, and overall educational quality across multiple universities during the pandemic. Similarly, Santoso et al. [30] demonstrated that 90% of academics at Tshwane University of Technology agreed that AI-powered LMS tools, such as discussion forums, chatbots, and assessment features, enhanced student participation and achievement in online learning environments. By providing such concrete examples of AI–LMS integration in real institutions [10,29], studies can better illustrate the tangible benefits and practical implications of these technologies, making a stronger case for their adoption and further research in the field.

Additionally, Berry [32] explored the application of AI to adaptive learning environments and found that institutions can implement AI-based adaptive e-learning systems in LMS that personalise instruction based on student learning style and knowledge level, leading to improvements in student knowledge levels and success. Similarly, Tendulkar et al. [33] investigated AI-based approaches for teaching and learning to develop critical thinking and problem-solving skills. They suggest that institutions can leverage AI applications in LMS to provide focused curriculum and learning paths for students to develop into independent learners with strong thinking and problem-solving abilities, enhancing educational quality and student success.

Moreover, institutions can develop AI-driven dynamic and personalised adaptive learning systems in LMS by analysing learner behaviour to enable more effective learning environments and improve overall institutional performance [34]. In a different vein, Paba-Medina et al. [29] designed a gamified tool for developing citizenship competencies. They suggest that institutions can integrate AI-powered gamification tools with LMS to reinforce the teaching and learning of important skills like citizenship competencies engagingly, contributing to enhanced educational quality and student success.

The findings from these studies provide valuable insights into how higher education institutions can effectively leverage the integration of AI and LMS to enhance educational quality, student success, and overall institutional performance (Table 5). Key strategies include using AI-enabled conversational agents and learning analytics, applying AI and data mining techniques to LMS data, extending LMS with AI-driven adaptivity, leveraging AI-based educational platforms and tools, encouraging the use of AI-powered tools in LMS, integrating AI into LMS for learning analytics, implementing AI-based adaptive e-learning

systems, using AI applications in LMS to provide focused curriculum and learning paths, developing AI-driven adaptive learning systems, and integrating AI-powered gamification tools with LMS.

Table 5. Research summary.

Authors and Year	Research Focus	Research Settings	Key Findings
Atif et al. [2]	Remote learning and teaching using pedagogical conversational bots and learning analytics utilising AI	Multiple universities	AI-enabled conversational agents and learning analytics in LMS may boost student engagement, personalised assistance, and empowerment, improving educational quality and achievement.
Kunnath et al. [10]	Data mining and AI-based adaptive learning management systems	Engineering college	LMS conversational agents and learning analytics may improve student engagement, empowerment, and educational quality and accomplishment.
Lim et al. [28]	Integrating learning styles and adaptive courses in LMS	University in Malaysia	Institutions may add AI-driven adaptivity to LMS based on student learning styles to provide personalised assistance.
Qushem et al. [3]	Precision education using multimodal technology	Multiple research examples	AI-based educational platforms and technologies show significant potential.
Santoso et al. [30]	Academic LMS digital technology student engagement abilities	Tshwane University of Technology	Universities may promote AI-powered LMS technologies like discussion forums, chatbots, and evaluation features to boost student engagement and achievement.
Sharif et al. [31]	An AI-powered learning analytics framework to improve student learning	Learning analytics to personalise learning	Universities may use AI in LMS for learning analytics to personalise learning, identify at-risk children, and provide instructors with data-driven insights to improve student education and institutional performance.
Tendulkar et al. [31]	AI-based critical thinking and problem-solving instruction	Concentrated curriculum and learning pathways	AI applications may concentrate curriculum and learning pathways to help students become autonomous learners with strong thinking and problem-solving abilities.
Zagorskis et al. [34]	Learning analytics architecture for LMS machine learning algorithm integration	Riga Technical University	Using learner behavioural data, institutions may create AI-driven dynamic and personalised adaptive learning systems in LMS to enhance learning environments and institutional performance.
Paba-Medina et al. [29]	Citizenship competency development using gamification	AI-powered gamification tools	Institutions may use AI-powered gamification tools with LMS to engage students in citizenship skills and improve educational quality.

5. Conclusions

The systematic literature review uncovers numerous advantages and possibilities linked to integrating AI and LMS in higher education institutions. Uunona & Goosen [19] discovered that online learning applications powered by AI can have a positive impact on diversity, inclusion, and student outcomes in a university setting. Additionally, Atif et al. (2021) [2] found that AI-enabled conversational agents and learning analytics in LMS had positive effects in a blended learning environment during the pandemic. They observed that these technologies provided personalised support, enhanced engagement and learning, and empowered students. The integration of AI with learning management systems (LMS) presents both opportunities and challenges for various stakeholders in higher education. For university administrators, AI-enabled tools such as conversational agents and learning analytics can enhance educational quality by personalising assistance and boosting student engagement and performance. Faculty members can benefit from AI-powered tools that encourage interactivity and engagement in online learning environments, as well as provide focused curricula to develop students' critical thinking and problem-solving skills. Students, on the other hand, can experience personalised support and enhanced learning through AI-enabled conversational agents and AR/VR technologies, which can also aid learners with physical challenges.

In addition, Hamza et al. [26] found that incorporating AI and AR/VR into LMS enhanced personalised learning methods and the development of life skills for learners with physical challenges. Also, AI applications in LMS can enhance personalised learning paths, offer adaptive assessments, and generate data-driven insights to enhance the educational process and improve learning outcomes. The results show the ability to combine AI and LMS to revolutionise and improve the learning experience for students from various backgrounds. However, integrating AI and LMS in higher education presents a range of obstacles and ethical considerations that must be acknowledged and resolved. These include the protection of data privacy and security, the need to address algorithmic bias and ensure fairness, the ethical considerations surrounding student autonomy and consent, the importance of bridging the digital divide and promoting accessibility, and the necessity of adequate faculty training and adoption. Qusheh et al. [3] also identified the importance of addressing ethical considerations and privacy concerns when implementing AI technologies in educational settings. In addition, the challenges of integrating AI into LMS include the importance of privacy-preserving measures and effectively managing diverse data [13]. The study also revealed a need for instructional design support in selecting appropriate AI-powered tools for the LMS. This is crucial to ensure that these tools are not only effective but also used ethically. The significance of addressing ethical and practical challenges cannot be overstated to ensure the successful and responsible integration of AI and LMS in higher education.

Also, the systematic literature review demonstrates that AI–LMS integration significantly contributes to sustainability in higher education through three key dimensions. First, environmental sustainability is enhanced through paperless learning environments and optimised resource utilisation, with institutions reporting measurable reductions in physical resource consumption. Second, social sustainability is strengthened through improved educational access and personalised learning pathways, supporting inclusive education goals, particularly for underserved populations. Third, economic sustainability is achieved through cost-effective delivery models and efficient resource allocation, though initial implementation costs must be carefully considered. The findings reveal that when thoughtfully implemented, AI-powered LMS solutions can simultaneously advance educational quality and institutional sustainability goals. However, challenges remain in ensuring equitable access to digital resources and managing the energy footprint of these technologies. Future research should focus on quantifying the long-term sustainability impacts of AI–LMS integration and developing implementation frameworks that balance technological advancement with sustainable development objectives.

Furthermore, findings revealed that, to maximise the benefits of combining AI and LMS, higher education institutions can implement a range of strategies that can improve the quality of education, enhance student success, and boost overall institutional performance. A study discovered that the integration of AI-enabled conversational agents and learning analytics in LMS has the potential to enhance educational quality and student performance. These technologies offer personalised assistance, increase engagement and learning, and empower students throughout their educational journey. In a similar vein, Kunnath et al. [10] showcased the potential for universities to leverage AI and data mining techniques. By analysing LMS data, they were able to identify areas where students lacked understanding and provide tailored learning paths to enhance learning efficiency and promote student success. Additionally, universities could improve their learning management systems (LMS) by integrating artificial intelligence (AI) technology. This AI technology would be able to adapt to students' individual learning styles, offering personalised support for both instructors and learners. The ultimate result would be enhanced learning effectiveness and an overall improvement in educational quality. Furthermore, results show that institutions can utilise AI-powered educational platforms and technologies such as MOOCs, games, LMS, AR/VR, and more to personalise learning experiences and enhance the acquisition of knowledge and skills. This approach has shown promising outcomes and has the potential to enhance the overall quality of education. The results emphasise the power of combining AI and LMS to revolutionise and enhance the learning process, resulting in improved educational outcomes and institutional performance.

6. Future of AI-Powered LMS in Higher Education

The potential impact of AI-powered learning management systems (LMS) in higher education is immense, offering the opportunity to transform teaching, learning, and administrative processes. Integrating AI with LMS will allow for customised learning experiences, advanced tutoring systems, and detailed learning analysis, leading to better student results and institutional performance. However, it is crucial to address ethical concerns such as data privacy, algorithmic bias, and faculty training for successful implementation. With the increasing demand for flexible and lifelong learning, AI-powered LMS will be crucial in meeting these needs by providing engaging and effective learning experiences. In the end, the potential of AI-powered LMS to revolutionise higher education is incredibly promising. It can create a more intelligent, adaptable, and inclusive ecosystem.

University administrators are tasked with navigating the complex landscape of online education, which presents both opportunities and challenges. The expansion of online learning can provide a means to generate new revenue streams and address the rising costs of higher education, particularly in developed countries. This can enable universities to broaden their reach and attract a more diverse student population, including those from underserved regions and backgrounds.

The focus of this study is the integration of AI and LMS in higher education settings. The narrow focus of this study restricts the ability to apply the results to different educational settings, such as K-12 schools or vocational training programmes. In addition, due to the rapid pace of technological advancements, certain discoveries may have limited relevance in the long run. There is a potential gap when relying solely on published research articles, as they may not include the most recent advancements and innovations in the field. In addition, it is important to consider the selection criteria and limitations of the databases used, which may have impacted the comprehensiveness of the review. It is worth mentioning that this study does not offer in-depth technical guidance or specific implementation recommendations. Further exploration is needed to fully understand the financial and infrastructural implications of integrating AI and LMS, which go beyond the scope of this review.

Funding: This research is self-funded, and no funds were used from the agencies and institutions. This approach enabled a thorough investigation into the integration of AI and LMS in higher education, contributing to the enhancement of educational quality and student success.

Data Availability Statement: The systematic literature review methodology is based on the PRISMA Statement 2020. The major purpose was to investigate how artificial intelligence and big data analytics may improve higher education. An exhaustive search for relevant research articles in the Scopus database was conducted at this stage of the inquiry. A keyword pattern was used to ensure a search that is especially focused on higher education in accordance with the study strategy. The predefined keyword query yielded 102 research publications that satisfied the given requirements. After examining the 102 research publications, it was determined that they were pertinent to the study subject. A further examination identified 82 publications that provided a response to the study inquiry. Following a comprehensive linguistic examination, a total of 60 documents were selected for further investigation.

Acknowledgments: I would like to extend my sincere gratitude to Imam Mohammad bin Saud Islamic University for providing the necessary resources and support throughout the research process. Special thanks to my colleagues at the College of Languages and Translation for their valuable insights and encouragement. Additionally, I would like to express my appreciation to all the researchers whose work has been referenced in this study, as their contributions have been instrumental in shaping this research. Finally, I would like to thank my family and friends for their unwavering support and understanding during this project.

Conflicts of Interest: The author declares no conflicts of interest.

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